DEEDY: A Thai-Centric Multi-Agent Framework for Social-Media Marketing Campaign Analysis

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Abstract*—Social-media marketing campaigns generate large volumes of fast-moving, context-dependent consumer reactions that are difficult to interpret with sentiment classification alone. This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), a Thai-centric application and experimental testbed for analyzing campaign response with large-language-model (LLM) agents. For each campaign post, an Apify-based pipeline requests up to 1,000 publicly accessible comments from Facebook, Instagram, or TikTok, preserves post and collection provenance, and produces synchronized raw and Thai-aware analysis views. Campaign facts and approved context form a reality seed, whereas observed reactions reserved for evaluation remain hidden from the agents. DEEDY adapts MiroFish, an open-source multi-agent prediction engine that converts seed documents into a GraphRAG-grounded world, generates actors, runs OASIS social interactions, and supports report generation and post-run inspection. The resulting application lets marketers examine how message framing, audience composition, and platform mechanisms may shape simulated reaction, diffusion, and narrative formation. We propose leakage-resistant evaluation against held-out Thai comments using sentiment and stance distribution distance, narrative coverage, response diversity, propagation error, repeated runs, and native-Thai human assessment. An OpenRouter pilot completed 500 calls across DeepSeek V4 Flash 0731 and Qwen3 8B, but its paired diagnostic intervals did not establish a grounding or model-fidelity advantage. DEEDY is positioned as an exploratory campaign-planning and social-listening instrument, not as a deterministic forecast of consumer behavior or a replacement for field research.

***Index Terms*—LLM agents, social-media marketing, campaign analysis, social simulation, Thai NLP, GraphRAG, MiroFish, Apify**

I. Introduction

Marketing campaigns on social media are not one-way messages. Consumers reply to a campaign post, react to one another, repeat or contest emerging claims, and carry interpretations across platform feeds. A favorable first response may be displaced by complaints about price, product quality, brand conduct, or the credibility of an influencer; conversely, a small positive narrative may grow through recommendation and peer endorsement. Counting likes or assigning sentiment to comments can summarize a snapshot, but it cannot by itself represent the interaction process through which campaign narratives emerge. Agent-based modeling (ABM) provides a bottom-up way to represent local exposure and response decisions and observe their collective consequences [1].

LLM agents extend ABM by allowing simulated actors to condition open-ended actions on a profile, memory, retrieved evidence, feed context, and earlier conversation. Generative Agents demonstrated memory-driven believable behavior [2], while Social Simulacra showed how generated users and conversations can support social-computing prototypes [3]. Systems including S^3, OASIS, and GenSim subsequently modeled social networks, recommendation, and multiple online actions at larger scales [4], [5], [6]. These capabilities suggest a campaign-analysis application in which marketers can rehearse alternative message framing, audience composition, and feed conditions before interpreting the resulting differences as hypotheses for further testing.

MiroFish provides the application foundation for this work. The open-source project describes itself as a multi-agent prediction engine that constructs a parallel digital world from real-world seed material such as news, policy drafts, or analysis reports [7]. Its workflow has five stages: graph building with seed extraction and GraphRAG memory injection; environment setup with entity, persona, and agent construction; dual-platform social simulation with temporal memory updates; report generation; and deep interaction with individual agents or the report agent. OASIS supplies the underlying social-simulation engine. MiroFish therefore offers a useful bridge from campaign documents to an inspectable simulated environment, but its outputs must still be validated before they are used to support marketing decisions.

Applying this workflow to Thai campaigns introduces two challenges. First, Thai localization is more than translating an English prompt. Thai has no obligatory whitespace between words, and campaign comments frequently combine informal spelling, repeated characters, emoji, hashtags, transliterated English, regional language, sarcasm, indirect disagreement, and culturally specific politeness. Thai NLP resources and models have improved substantially [8], [9], [10], [11], but cultural capability remains a distinct evaluation target [12]. Second, social data collection is platform-dependent and incomplete. A reproducible study must record which campaign post was queried, the collection time, the requested limit, the returned count, and any access or visibility limitation.

A further methodological problem is that plausible campaign comments do not prove behavioral fidelity. Demographically conditioned LLM samples can approximate some human distributions [13], yet recent work finds behavioral inconsistency across settings [14]. Operational validation against authentic comments also shows that matching surface form can be separated from matching content [15]. For this reason, campaign context needed to construct the simulation must be separated from real consumer reactions used to test it, and stochastic conditions must be repeated rather than judged from a single persuasive-looking run.

This paper presents **DEEDY**, a Thai-centric campaign-analysis application that combines an Apify data-preparation pipeline, the MiroFish core, and an explicit validation layer. Analysts provide campaign topics and public post URLs from Facebook, Instagram, or TikTok. DEEDY requests up to 1,000 comments for each post, preserves provenance and conversation structure, and separates campaign context from held-out consumer reactions. Within the simulator, an analyst can compare a baseline campaign with counterfactual message framing, actor composition, network structure, or recommendation policy. The application reports where simulated and observed discourse agree or diverge; it does not claim to forecast the choices of a specific person or the Thai market as a whole.

Our contributions are threefold:

- We define an end-to-end Thai campaign-data pipeline that uses Apify Actors to request up to 1,000 publicly accessible comments per campaign post, normalize cross-platform fields, preserve provenance, and produce Thai-aware views for analysis and simulation.
- We adapt the MiroFish/OASIS workflow to social-media marketing through campaign-grounded actor construction, Thai linguistic attributes, configurable campaign interventions, and auditable simulation outputs.
- We specify a leakage-resistant protocol that evaluates content, population, and dynamic correspondence against held-out Thai campaign reactions using repeated runs, native-Thai review, ablations, and uncertainty reporting.

II. Related Work

A. Generative Agents and Social Simulation

Traditional ABMs encode actor states and behavioral rules explicitly. LLM agents complement this approach by generating context-dependent language and actions, but introduce sensitivity to prompts, decoding, memory, and the shared base model. Generative Agents use observation, reflection, and planning to produce coherent-seeming routines [2]. S^3 connects LLM perception and action to emotion, attitude, and a social network [4]. OASIS adds dynamic user and post state, follow/comment/repost actions, and configurable interest- and popularity-based recommendation at large agent scales [5]. GenSim emphasizes reusable scenario functions and execution robustness [6]. DEEDY uses these systems as infrastructure, but focuses on Thai campaign construction and empirical validation rather than maximum population size.

B. Grounding and Operational Validity

MiroFish extracts entities and relations from uploaded reality seeds, stores individual and collective memory in a graph-retrieval layer, generates OASIS profiles, runs social platforms, and uses a report agent to query simulation state [7]. This workflow accelerates scenario construction, but the generated report is still evidence about the simulated world, not evidence that the world is valid. DEEDY therefore treats every simulation trace as a hypothesis to compare with unseen observations.

Algorithmic fidelity compares LLM-generated and human subgroup distributions [13]. Conditioned Comment Prediction instead compares a generated response with the authentic response a user left to a stimulus, providing a closer operational test for social-media simulation [15]. Validation of generative ABMs further benefits from staged replication before novel counterfactual tests [16]. These ideas motivate DEEDY's campaign-level holdout: material required to describe the campaign is available to agents, whereas the consumer reactions used for scoring are not.

C. Thai Online Language and Campaign Response

Thai NLP remains comparatively resource-constrained, especially for informal and downstream semantic tasks [8]. PyThaiNLP provides tokenization and other open Thai processing components [9]; WangchanBERTa showed that Thai-specific preprocessing and pretraining over news and social data matter [10]; and Typhoon 2 extends Thai-optimized models to instruction-following generation [11]. ThaiCLI separately evaluates cultural and linguistic knowledge, highlighting that general benchmark strength does not guarantee Thai cultural competence [12].

Online affect is also contextual. Sarcasm can invert literal polarity, and conversation history and user style improve its detection [17]. This is important in Thai discussions, where particles, quotation, exaggerated praise, wordplay, and shared cultural references may signal stance indirectly. Accordingly, DEEDY keeps thread context and raw text for evaluation and does not treat sentiment, stance, and sarcasm as interchangeable labels.

III. DEEDY Framework

A. System Architecture Overview

Figure 1 organizes DEEDY into three layers. The *data layer* selects campaign posts, collects comments through Apify, cleans, partitions, and annotates Thai campaign material. The *core layer* supplies the scenario summary to MiroFish, which constructs a GraphRAG-backed world and executes the hybrid simulation layer adapted from the MarketFish component design. This layer combines cognitive LLM agents with stochastic public agents, dual memory, FYP or KOL seeding, network propagation, and an optional information-operations stress test. The *validation layer* compares outputs with social posts and comments never shown to the agents. Every derived artifact records a campaign identifier, platform, post URL, provenance pointer, processing version, and experimental split so that a score can be traced back to its evidence.

B. Data Preparation Pipeline

1) Campaign Definition and Post Selection: The input manifest `data/topic_ref.json` defines one marketing campaign per record through a campaign topic and a list of reference URLs. URLs may refer to campaign posts, creator

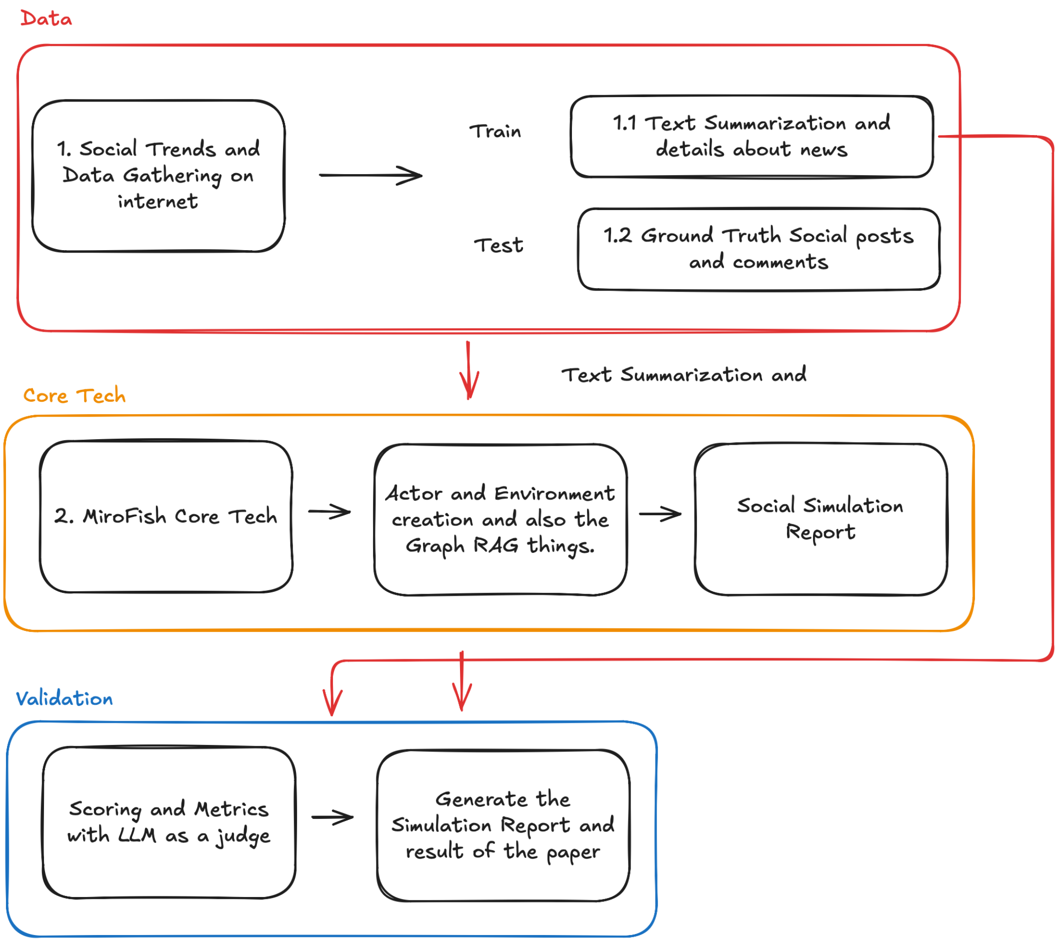

Fig. 1. DEEDY application architecture. Thai campaign context and social-media data are separated into a scenario-construction stream and a held-out validation stream. The MiroFish core combines graph grounding, dual memory, a hybrid cognitive–stochastic population, configurable exposure mechanisms, and auditable outputs for multi-level evaluation.

or influencer posts, news coverage, or public community posts on Facebook, Instagram, and TikTok. Including more than the brand's official account reduces the risk of measuring only an owned-audience response. For an experimental release, the manifest is extended with a stable campaign identifier, observation window, and source stratum. Each URL is manually checked for campaign relevance and resolved to a canonical post before collection. The sampling frame and inclusion rules are frozen before labels or simulation outputs are inspected.

Campaign facts used to construct the simulated scenario are stored separately from consumer reactions. The factual packet may include the campaign brief, creative copy, product information, publication time, and dated public reporting. It does not include the target comments later used for evaluation. This distinction supports both retrospective campaign analysis and a stricter prospective design in which only information available at campaign launch is provided to the agents.

2) Apify Comment Collection: DEEDY uses Apify Actors as the collection interface rather than a manually maintained browser crawler. An Actor is a server-side program that accepts structured input, executes as a recorded run, and writes results to an Apify dataset accessible through the API or client libraries [18]. The implemented `apify_crawler.py` detects the platform from each URL and invokes `apify/facebook-comments-scraper`, `apify/instagram-comment-scraper`, or `clockworks/tiktok-comments-scraper`. A platform adapter maps each Actor's output into one common comment schema.

The collection unit is one social-media post. Under the revised protocol, DEEDY launches an independent Actor query for every canonical post URL with a requested limit of **1,000 comments per post**. The limit is reset for each URL; it is not a shared 1,000-comment budget for the campaign or topic. If a campaign contains m posts, the intended maximum is therefore

$$N_{\max} = 1{,}000m. \tag{1}$$

The returned count may be lower when a post has fewer comments or when the platform or Actor cannot expose every comment. Replies are configured explicitly for each platform and are counted as separate records when returned. Accordingly, "1,000" is a reproducible query cap, not a guarantee of 1,000 observations and not a claim of complete platform coverage.

3) Normalization and Provenance: Each returned item is validated and normalized to the fields `topic`, `platform`, `post_url`, `comment_id`, anonymized `author`, `text`, `likes_count`, `parent_comment_id`, and `timestamp`. The parent identifier preserves reply structure when the Actor exposes it. Missing values remain explicit rather than being inferred. Records without comment text are excluded from content analysis but retained in run-level quality-control counts when they represent Actor errors.

For reproducibility, a provenance manifest records the campaign and canonical post identifiers, Actor identifier and version, complete run input, Apify run and dataset identifiers, start and completion times, requested limit (1,000), returned count, reply setting, error status, collection-code version, and a checksum of the frozen raw export. Raw UTF-8 JSONL is immutable after a successful run. Subsequent tables are derived from that snapshot and de-duplicated first by platform comment identifier and then by normalized-text similarity within a post. User names and platform identifiers are hashed or removed before research analysis.

This design improves repeatability but does not turn the result into a random sample. Search and post selection, platform ranking, deleted or moderated comments, public visibility, geography, login state, and Actor behavior can all affect what is returned. DEEDY therefore reports requested and observed counts for every post and treats cross-platform comparisons as conditional on these collection mechanisms.

4) Thai-Aware Data Views: The crawler preserves the Unicode comment text returned by each Actor in an immutable JSONL view. The implemented `convert_comments_to_csv.py` then flattens nested post records into a UTF-8-SIG table containing the campaign topic, comment, author, engagement, platform, post, comment and parent identifiers, and timestamp. This second view supports inspection and annotation without replacing the raw export.

DEEDY avoids aggressive text cleaning because emoji, Thai–English code-mixing, orthographic repetition, hashtags, particles, negation, and punctuation may carry campaign stance or affect. The analysis view standardizes only encoding, line breaks, and whitespace; any additional transformation is versioned and never overwrites the raw text. Thai tokenization is performed only when a metric requires tokens, while character- and embedding-level metrics are reported in parallel to reduce dependence on one segmentation system.

5) Evidence Split and Structured Weak Annotation: For each campaign e, the collected evidence is partitioned before any model summary is generated:

$$D_e = S_e \,\dot{\cup}\, G_e, \tag{2}$$

where S_e is the scenario-construction set and G_e is the held-out observed reaction set. S_e contains the campaign facts and context permitted at the chosen simulation start time. G_e contains comments and interactions that the simulator must reproduce but never receives as memory. A post is assigned wholly to one experimental role when post-level holdout is used; temporal experiments instead apply a pre-registered timestamp cut. Comment identifiers, quoted passages, and near duplicates cannot cross the boundary.

The frozen Apify records are passed to `llm_prelabel.py` in bounded batches. Three critique models independently assign a positive, neutral, or negative label to each sampled comment and summarize campaign response. A fourth model synthesizes their outputs into consensus sentiment labels, an overall sentiment-direction analysis, recommended campaign next steps, and possible business or operational changes. Exact sentiment counts are computed from the consolidated comment-level labels rather than estimated from the synthesis text. Additional evaluation labels for campaign stance, narrative, purchase intent, and sarcasm are collected under the human-annotation protocol rather than attributed to the current pre-labeling script.

These model outputs are *weak pre-annotations*, not ground truth. Only a summary derived from S_e may enter MiroFish; labels or summaries derived from G_e are restricted to scoring and error analysis. Before the main experiment, native Thai annotators review a frozen, source-stratified sample, adjudicate disagreements, report Krippendorff's α, and audit errors involving sarcasm, code-mixing, indirect criticism, and platform source. This separation prevents observed campaign reactions from leaking into agent memory.

C. MiroFish Core Engine

1) Reality Seed, GraphRAG, and Dual Memory: The scenario packet contains the campaign brief, creative message, dated factual claims, named entities, brands, products, creators, and source-level provenance from S_e. MiroFish extracts an ontology and entity relations, then injects individual and collective memories into its graph retrieval layer [7]. A retrieved memory attached to an agent action includes its supporting node or source identifier. Unsupported details generated during graph construction are marked as model assumptions and are not silently promoted to facts.

The component design adds two timescales of per-agent memory. A short-term store holds exposures, private-state updates, and interactions within the active simulation window; a persistent ChromaDB store holds vectorized prior and reflected memories retrieved across rounds. This division reduces the "cold start" behavior of blank-slate agents without exposing them to the evaluation set: only material from S_e, together with events generated inside the simulation, may enter either store. Retrieval records the memory identifier, source class, similarity score, and simulated timestamp. The research prototype exposes a Redis-compatible short-term interface but currently uses an in-process substitute, so Redis latency and durability are not claimed as evaluated results.

2) Actor and Environment Construction: Actors represent campaign-relevant audience roles justified by S_e, such as existing customers, prospective customers, creator followers, brand advocates, skeptics, journalists, community members, and bystanders. A profile separates evidence-backed attributes from sampled attributes. The Thai adaptation adds language register, Thai–English mixing rate, region or dialect only when supported, politeness strategy, typical message length, activity rate, campaign-interest prior, interests, and uncertainty. Demographic attributes are not inferred from a single message. Population weights and network links are sampled from documented priors and varied during sensitivity analysis rather than presented as known facts.

The environment specifies platforms, action space, time step, feed capacity, and recommendation mechanism. OASIS supplies posting, replying, reposting, following, and browsing behavior with dynamic social and content state [5]. DEEDY additionally records each exposure before an action. This makes it possible to distinguish "the agent never saw the message" from "the agent saw and rejected the message," a necessary distinction for interpreting diffusion failure.

3) Hybrid Cognitive–Stochastic Population: Directly invoking an LLM for every decision limits population size, increases cost, and makes a viral exposure burst vulnerable to rate limits. DEEDY therefore integrates the component paper's hybrid ABM design. Let $\mathcal{A} = \mathcal{C} \dot\cup \mathcal{M}$ be the agent population, where $\mathcal{C}$ contains cognitive agents and $\mathcal{M}$ contains stochastic mathematical agents. The cognitive tier is a small, configurable share of the population—5% in the prototype configuration, rather than a universal constant—and is assigned to influential or analytically important roles. These agents retrieve memory, update private stance, generate context-aware Thai language, and choose among platform actions. The remaining public agents retain a profile, prior bias, network position, and exposure history, but use a lightweight probabilistic response rule.

For an exposed mathematical agent i at round t, the probability of taking a visible action rather than abstaining is represented as

$$\Pr(a_{i,t} \neq \varnothing) = \sigma\Big(\beta_0 + \beta_b b_i + \boldsymbol{\beta}_d^\top \mathbf{d}_i + \beta_s s_{t-1} + \beta_x x_{i,t}\Big), \tag{3}$$

where σ is the logistic function, b_i is the agent's pre-campaign bias, $\mathbf{d}_i$ contains documented population attributes, s_{t-1} is the preceding visible sentiment trend, and $x_{i,t}$ contains exposure and neighbor signals. A second categorical draw selects like, share, comment, or silence. All coefficients, tier ratios, and random seeds are versioned per run. The model does not assume that public opinion is determined by KOL output: tier-specific results are logged, and sensitivity analysis varies both the cognitive share and β_s to reveal artificial herding.

4) Algorithmic Seeding and Network Propagation: The initial exposure mechanism represents discovery outside a follower graph. Under FYP seeding, round 1 places the campaign in the observation queues of a configured sample of eligible agents. Under KOL injection, the same campaign is delivered first to selected high-influence cognitive agents. Later rounds propagate shares over a configurable small-world or observed interaction graph. The exposure probability is

$$p_{i,t}^{\text{see}} = \text{clip}_{[0,1]}\left[\alpha f_{i,t} + (1-\alpha)\sum_{j\in\mathcal{N}(i)} w_{j,i} I_{j,t-1}^{\text{share}}\right], \tag{4}$$

where $f_{i,t}$ is algorithmic eligibility or targeting, $\mathcal{N}(i)$ is the neighbor set, $w_{j,i}$ is directed influence, and $I_{j,t-1}^{\text{share}}$ indicates a share in the previous round. When the prior wave's engagement exceeds a pre-registered threshold τ, the FYP mechanism may open a secondary wave to unreached strata. The seed size, α, τ, edge construction, and eligibility policy are experimental assumptions, not reconstructions of a platform's proprietary recommender. Chronological, popularity, and interest-based policies remain comparison conditions.

5) Thai Context in Perception and Action: At each round, an actor retrieves relevant graph memory, observes a bounded feed, updates a private state, and chooses an action or abstention. Prompts require natural Thai consistent with the actor's register without demanding caricatured dialect. They preserve uncertainty, allow silence, and prohibit fabricated quotations or demographic claims. Conversation context is included for reply generation because Thai stance and sarcasm may depend on an earlier turn [17]. The language model is configurable: a Thai-optimized model such as Typhoon 2 can be compared with multilingual models under identical action rules rather than assuming that any model is superior. In the executable pilot, the OpenAI-compatible OpenRouter endpoint was configured with pinned `deepseek/deepseek-v4-flash-0731` and `qwen/qwen3-8b` revisions. Neither is treated as a Thai-specialized control; their comparison tests the configured model path and cost–quality trade-off, while the native-Thai model comparison remains future work.

6) Information-Operations Stress Testing: The optional information-operations (IO) module assigns a pre-registered share of mathematical agents to coordinated positive promotion or negative attack. Their injected actions are explicitly labeled and kept separate from organic responses, allowing the analyst to measure changes in reach, sentiment, polarization, and cross-group adoption as the IO rate varies. This module is a counterfactual robustness test, not a detector of real accounts and not a claim that authentic Thai users follow bot behavior. It is disabled in the primary correspondence experiment unless IO prevalence is itself the registered factor.

7) Simulation Outputs and Application Layer: An asynchronous FastAPI orchestration service accepts simulation configuration through REST, bounds concurrent LLM requests, and publishes tick, action, network, and aggregate updates to the React application through WebSockets. This permits a live matrix feed without making UI delivery part of the agent's decision rule. The core also returns a machine-readable simulation log and a human-readable campaign report. The log contains agent state version, exposure set, retrieved evidence, selected action, generated content, parent action, timestamp, and token/model settings. The report summarizes dominant and minority narratives, sentiment and stance trajectories, campaign reach, engagement mix, influential nodes, propagation paths, disagreement, and unsupported claims. Live sentiment timelines and reach distinguish exposed-and-engaged agents from exposed-but-silent agents; unreached agents remain a third category. The report also exposes uncertainty across runs and separates cognitive, mathematical, and injected IO activity.

The application allows a marketer or analyst to select a campaign, inspect the scenario packet and graph, configure actor count, hybrid-tier ratio, seed policy, feed policy, and IO mode, run a baseline or a counterfactual campaign message, compare it with held-out observations, and drill down from an aggregate chart to the simulated posts and evidence that produced it. Counterfactual reports compare conditions inside the model; they are not presented as causal effects in the real Thai consumer population.

IV. Experimental Design

A. Research Questions

The evaluation is organized around four questions:

- **RQ1 (Content):** Does DEEDY reproduce the sentiment, stance, narrative, and linguistic characteristics of held-out Thai reactions?
- **RQ2 (Population):** Does graph grounding and Thai adaptation improve aggregate correspondence without collapsing response diversity?
- **RQ3 (Dynamics):** Does the simulated interaction process match observed cascade size, shape, timing, and network structure?
- **RQ4 (Application):** Are campaign reports traceable, stable across repeated runs, and useful to Thai marketing analysts without overstating certainty?

B. Dataset and Splitting Protocol

The initial study will select campaigns with at least one usable factual scenario packet and multiple public campaign-related posts. Before simulation, the protocol freezes the required minimum returned-comment count, eligible platforms, post-source strata, and missing-data rules. Apify then receives a separate query for each post with a 1,000-comment limit as described in Section III-B2. Posts returning fewer records are retained when they satisfy the pre-registered minimum; the dataset never pads a short thread by duplicating comments or borrowing another post's quota.

The unit of generalization is a *campaign*, not an individual post or comment. Prompt and model selection use development campaigns, and final scores are reported on untouched test campaigns. Within a campaign, S_e and G_e are separated either by a pre-registered timestamp or by whole post so that a reaction thread cannot leak into both construction and evaluation. Actor run inputs, query dates, returned counts, source strata, and split assignments are frozen before weak annotation. This design supports source-balanced analysis, temporal-drift checks, and sensitivity to platform-specific public visibility.

Weak labels are independently reviewed by at least three native Thai annotators using a codebook for three-way sentiment, campaign-specific stance, sarcasm, and narrative membership. Annotators see necessary conversational context but not whether a post is real or simulated during quality comparison. Majority/adjudicated labels form the reference set, and Krippendorff's α is reported for each task. User identifiers are hashed and quoted examples are paraphrased unless publication is ethically justified.

C. Metrics

1) Content and Population Correspondence: Let P_G^y and P_S^y be observed and simulated distributions for label $y \in \{\text{sentiment}, \text{stance}, \text{narrative}\}$. Distributional correspondence is measured with Jensen–Shannon distance:

$$d_{\text{JS}}(P_G^y, P_S^y) = \sqrt{\tfrac{1}{2}D_{\text{KL}}(P_G^y \| M) + \tfrac{1}{2}D_{\text{KL}}(P_S^y \| M)}, \quad (5)$$

where $M = (P_G^y + P_S^y)/2$. Lower is better and the square-root form is bounded. Macro-F1 is also reported when simulated posts are matched to a conditioned stimulus or stance target.

Narrative correspondence is measured in two ways. First, Thai annotators map posts to a campaign-specific narrative codebook and we compare coverage and prevalence. Second, multilingual BERTScore measures semantic similarity between generated and observed response sets [19]. Embedding-based scores are treated as complementary because semantic proximity need not imply the same stance. Diversity is reported using Distinct-1/2, Self-BLEU, unique narrative count, lexical entropy, length distribution, and pairwise embedding distance. High correspondence accompanied by diversity collapse is considered a failure.

Polarization is measured from the stance distribution and interaction graph. We report normalized stance entropy, between-group versus within-group interaction, and modularity after fixing the community-detection procedure. Population statistics are computed per run and then aggregated across campaigns, so campaigns with many selected posts do not automatically dominate the result.

2) Temporal and Network Correspondence: For trace-enriched campaigns, cumulative activity curves are aligned to the campaign-post publication time. We report normalized root-mean-square error (NRMSE) and dynamic time warping distance, together with cascade scale, maximum depth, maximum breadth, structural virality, time to 50% of final activity, and peak-time error. OASIS uses scale, depth, breadth, and propagation-curve error in its own validation [5]; DEEDY adds exposure-conditioned action rates and Thai content labels.

Observed and simulated interaction graphs are compared by degree and cascade- size distributions using the two-sample Kolmogorov–Smirnov statistic, plus clustering coefficient, reciprocity, assortativity, modularity, and the share of cross-stance replies. These metrics are omitted, rather than imputed, when the source platform does not expose the required trace.

3) Thai Quality and Report Evaluation: Native Thai raters score blinded real and simulated posts on a five-point scale for naturalness, contextual coherence, stance consistency, pragmatic and cultural appropriateness, and persona consistency. Sarcasm precision, recall, and F1 are evaluated separately. Report claims are scored for evidence traceability: the proportion of factual claims linked to scenario evidence or simulation events, unsupported-claim rate, contradiction rate, and coverage of minority narratives.

TABLE I
PLANNED EXPERIMENTS. EACH CONDITION IS RUN WITH THE SAME CAMPAIGN SPLIT, AGENT BUDGET, ROUND BUDGET, AND TEN RANDOM SEEDS UNLESS THAT FACTOR IS VARIED.

Experiment	Conditions	Primary purpose
Exp. 1: Grounding	DEEDY seed-only; ungrounded LLM agents; deliberately leaky all-source upper bound	Test whether held-out correspondence comes from the scenario pipeline rather than parametric knowledge or target leakage.
Exp. 2: Thai adaptation	Full Thai adaptation; translated/unadapted MiroFish; normalization without context features	Measure sentiment/stance distance, narrative coverage, Thai quality, sarcasm, code-mixing, and diversity.
Exp. 3: Core model	DeepSeek V4 Flash 0731; Qwen3 8B; simple non-generative ABM baseline in the pilot, followed by a matched Thai-specialized comparison in the confirmatory study	Separate the contribution of the language model from the graph, population, and platform mechanisms.
Exp. 4: Social mechanism	Chronological, popularity, and interest feeds; observed versus random network; multiple population sizes	Test cascade, polarization, herding, and sensitivity to platform assumptions.
Exp. 5: Application	Baseline campaign versus one pre-registered message-framing counterfactual; report with versus without uncertainty/evidence links	Assess traceability, analyst usefulness, and whether presentation changes trust.

An LLM judge may assist high-volume screening, but cannot be the sole quality measure. Pair order is swapped, output identity is hidden, the judge must cite the supplied evidence before scoring, and its agreement with Thai raters is reported. These controls address known position bias in LLM-as-a-judge evaluation [20].

D. Execution and Statistical Analysis

All stochastic conditions use ten seeds initially; a power analysis on pilot variance determines the final number. Model name and revision, prompts, temperature, top-p, agent count, network seed, recommendation parameters, round count, stopping rule, and token cost are frozen before the test set is opened. We report medians and 95% bootstrap confidence intervals across campaigns. Paired conditions use a campaign-level permutation test or Wilcoxon signed-rank test with Holm correction. Effect sizes accompany p-values.

The OpenRouter pilot fixes temperature at 0.8, the completion ceiling at 2,200 tokens, reasoning mode off, concurrency at 12, and ten requested seeds. Each call asks for 12 structured synthetic Thai reactions. Five campaigns, five prompt conditions, two models, and ten seeds yield 500 unique calls. The seed-only condition is reused when expanding Experiments 1–3, producing 750 metric rows without making duplicate provider requests. Model-returned sentiment labels are diagnostic self-reports rather than independent annotations.

The main comparison is the full DEEDY pipeline versus translated/unadapted MiroFish. Component ablations remove graph retrieval, Thai context features, thread context, or observed population weights one at a time. The deliberately leaky condition in Table I is never a valid system result; it is a diagnostic ceiling showing how much apparent performance can be gained by exposing held-out reactions.

E. Acceptance Criteria for the Application

Before results are observed, the study will set minimum criteria: (i) full pipeline completion and provenance for every scored run; (ii) statistically lower sentiment/stance Jensen–Shannon distance than the unadapted baseline; (iii) no significant diversity collapse relative to held-out posts; (iv) Thai quality above the midpoint with acceptable annotator agreement; (v) report unsupported-claim rate below a pre-registered threshold; and (vi), for trace-enriched campaigns, improvement on at least two cascade metrics without material degradation on the others. Failure remains a reportable result and defines the application's operating boundary.

V. AUTOMATED PILOT RESULTS

A. Execution Scope and Reproducibility

An automated pilot was executed on 28 August 2026 to verify that the revised MiroFish application, non-generative ABM baseline, parameter sweep, provenance capture, and report path run end to end. The Docker image was built locally after adding the missing FastAPI runtime dependencies and switching the launch path to the current `frontend-v2` application. With cloud credentials absent during the first run, the web shell remained available while LLM- and Zep-dependent operations remained disabled. The backend health endpoint and frontend both returned HTTP 200 during the recorded run. A second Dockerized run on 29 August 2026 (Bangkok time) exercised the OpenRouter LLM path for Experiments 1–3. Zep-dependent graph-memory operations were not required for this bounded prompt-level pilot.

The pilot used seeds 1–10, 12 rounds, requested mean degree 6, and populations of 200, 500, and 1,000. Experiment 4 comprised 900 runs (5 campaigns $\times 3$ feed policies $\times 2$ network assumptions $\times 3$ population sizes $\times 10$ seeds). The automated portion of Experiment 5 added 100 runs (5 campaigns $\times 2$ message conditions $\times 10$ seeds) at $N = 500$. Experiments 1–3 added 500 unique OpenRouter calls and 750 expanded metric rows. Input SHA-256 hashes, parameters, run-level records, aggregate tables, smoke-test responses, provider pricing, and interpretation limits are stored in `MiroFish_App/result/manifest.json`, `llm_exp123_manifest.json`, and the accompanying machine-readable files. No authentic comment text, author identifier, or post URL was placed in the result bundle or sent to OpenRouter.

Table II distinguishes executed work from the evaluation still required for the paper's main claims. In particular, this pilot does not substitute model self-labels or character ratios for independent held-out labels, native-Thai ratings, or analyst judgments.

B. Dataset and Trace Audit

The supplied aggregate file and crawl contained five campaigns and exactly 5,000 comments from 33 posts. Each campaign contributed 1,000 comments. The campaign-specific unique-author counts ranged from 946 to 981. However, all 5,000 records had a null `parent_comment_id`. An observed reply graph and empirical cascade tree therefore could not be reconstructed. In accordance with the missing-trace rule in the experimental design, the pilot does not report observed cascade correspondence. For the mechanism sensitivity test, the planned observed-network condition was replaced by an explicitly labeled *post-affiliation proxy*: agents sampled from the same source-post stratum received local ties. Its comparator was a random graph with the same number of edges. Neither graph should be described as an observed Thai interaction network.

The sentiment distribution used to initialize each campaign was the summary over all collected comments. Consequently, the Jensen–Shannon distance in this section measures drift from a calibration target, not accuracy on held-out reactions. This circular calibration would be invalid as evidence for RQ1 or RQ2; it is retained only to diagnose whether the social mechanisms destabilize the supplied aggregate proportions.

C. OpenRouter LLM Diagnostics: Experiments 1–3

All 500 provider calls completed. DeepSeek returned 2,983 of 3,000 requested reactions; Qwen returned 3,000 of 3,000. Across all five prompt conditions, DeepSeek's 250 calls averaged 35.51 seconds and cost \$0.0276, whereas Qwen's 250 calls averaged 15.17 seconds and cost \$0.1109. The complete run cost \$0.1386. Thus, Qwen was faster in this execution, but DeepSeek was about four times less expensive. These totals depend on the provider route, token counts, and prices recorded in the manifest and are not universal model benchmarks.

Table III reports sentiment JSD and a character-level Thai ratio. In Experiment 1, the paired seed-only-minus-ungrounded JSD was -0.0063 for DeepSeek (empirical 2.5–97.5 percentile interval -0.1483 to 0.1756) and 0.0471 for Qwen (-0.3313 to 0.3484). Both intervals span zero. The leaky condition had lower mean JSD, but this expected result is invalid as system performance because target aggregates were supplied to its prompt.

For Experiment 2, Thai-context-minus-translated JSD was -0.0032 for DeepSeek (-0.1444 to 0.1344) and 0.0337 for Qwen (-0.1988 to 0.3302). The Thai-context-minus-normalization intervals also spanned zero. Thai-context responses were longer than normalization-only responses for 49 of 50 DeepSeek pairs and all 50 Qwen pairs, but had lower character-bigram diversity on average. These prompt proxies therefore do not establish that Thai adaptation improved quality. Thai-character ratios of 0.956–0.998 show script use, not naturalness, cultural appropriateness, sarcasm fidelity, or code-mixing quality.

Table IV compares the shared seed-only condition with the uniform non-generative baseline. DeepSeek-minus-Qwen JSD was -0.0287 (-0.3592 to 0.2179). DeepSeek-minus-ABM was -0.0207 (-0.2341 to 0.1955), and Qwen-minus-ABM was 0.0081 (-0.3765 to 0.4340). All intervals span zero, so this pilot establishes no fidelity advantage for an LLM tier or for either LLM. Qwen produced every requested schema item; the DeepSeek seed-only rows were 0.9833 schema-valid and contained no exact duplicates. Blinded native-Thai scoring and independent campaign holdouts are still required.

D. Social-Mechanism Sensitivity

Table V reports the $N = 500$ slice, averaging five campaigns and ten seeds (50 runs per row). Results were strongly sensitive to the assumed graph. The edge-matched random network produced mean action rates of 0.968–0.970 and reach of 0.997, whereas the post-affiliation proxy produced action rates of 0.400–0.402 and reach of 0.461–0.463. This difference is a property of the simulated propagation paths and is not an estimate of a platform effect. Population-size sensitivity was smaller within each topology: across the full sweep, mean action rates ranged from 0.965 to 0.970 for the random graph and from 0.400 to 0.419 for the post-affiliation proxy.

The chronological feed had the lowest calibration drift at $N = 500$ ($d_{\mathrm{JS}} = 0.077$ on the random graph and 0.082 on the proxy), followed by the interest feed (0.102 and 0.097) and popularity feed (0.120 and 0.115). Popularity also produced lower polarization in the random-graph condition. These are descriptive sensitivity results from the non-generative rule set; they do not establish that a real chronological feed is more accurate or less polarizing.

E. Conditional Message-Frame Smoke Test

The application pilot compared the baseline with a positive-clarity signal of 0.8 inside the model, using the interest feed, post-affiliation proxy, and $N = 500$. Across 50 campaign-seed pairs, the positive-frame condition increased mean positive share from 0.443 to 0.667 and mean sentiment from 0.306 to 0.615. The paired positive-share change was 0.225, with a 2.5–97.5 percentile interval of 0.106–0.364 across the 50 pairs. Action rate changed by only 0.017, and its empirical interval (-0.039 to 0.085) included zero.

The same intervention increased mean calibration JSD from 0.097 to 0.266; the paired change was 0.169 (empirical interval 0.054–0.291). Thus, the smoke test shows that a counterfactual travels through the pipeline and materially changes internal outputs, but it does not show an improvement. Indeed, it moves the simulation farther from the distribution used for calibration. Because the signal and response rule are model assumptions, these changes are conditional model behavior rather than causal effects on Thai consumers.

Overall, the automated pilot satisfies the engineering portion of application completion and provenance and exercises the

TABLE II
COVERAGE OF THE PLANNED EXPERIMENTS IN THE AUTOMATED PILOT.

Experiment	Pilot status
Exp. 1: Grounding	Seed-only, ungrounded, and deliberately leaky prompt conditions were run for two models; independent held-out scenario packets remain unavailable.
Exp. 2: Thai adaptation	Three prompt-level conditions were run for two models; native-Thai reference ratings remain unavailable.
Exp. 3: Core model	DeepSeek V4 Flash 0731, Qwen3 8B, and the simple non-generative ABM were run; the Thai-specialized and full hybrid-tier comparison remains pending.
Exp. 4: Social mechanism	The full automated feed, network-assumption, and population-size sensitivity sweep was run with ten seeds.
Exp. 5: Application	Docker health, result provenance, and a paired internal message-frame counterfactual were run; analyst usefulness and presentation effects were not rated.

TABLE III
EXPERIMENTS 1–2 OPENROUTER DIAGNOSTICS. EACH CELL AVERAGES FIVE CAMPAIGNS AND TEN SEEDS. SENTIMENT JSD IS DISTANCE FROM AN ALL-COMMENT CALIBRATION TARGET, NOT HELD-OUT ACCURACY; THE LEAKY ROW IS INVALID BY DESIGN.

Experiment	Condition	DeepSeek JSD	Qwen JSD	DeepSeek Thai ratio	Qwen Thai ratio
Exp. 1	Seed only	0.2572	0.2859	0.9822	0.9922
Exp. 1	Ungrounded	0.2635	0.2388	0.9734	0.9561
Exp. 1	Leaky invalid bound	0.2031	0.1966	0.9894	0.9927
Exp. 2	Thai-context prompt	0.2572	0.2859	0.9822	0.9922
Exp. 2	Translated/unadapted	0.2604	0.2523	0.9978	0.9611
Exp. 2	Normalization, no context	0.2460	0.2548	0.9797	0.9736

TABLE IV
EXPERIMENT 3 CORE-MODEL DIAGNOSTIC OVER FIVE CAMPAIGNS AND TEN SEEDS. COST IS MEAN PROVIDER COST PER CAMPAIGN-SEED ROW. THE ABM EMITS STATES, NOT LANGUAGE.

Condition	Sentiment JSD	Thai ratio	Valid share	Latency (s)	Cost/run
DeepSeek V4 Flash 0731	0.2572	0.9822	0.9833	31.49	$0.000116
Qwen3 8B	0.2859	0.9922	1.0000	16.53	$0.000476
Uniform non-generative ABM	0.2779	–	1.0000	0.00	$0.000000

TABLE V
EXPERIMENT 4 AUTOMATED PILOT AT $N = 500$. ENTRIES ARE MEANS OVER FIVE CAMPAIGNS AND TEN SEEDS. JSD IS DISTANCE FROM THE ALL-COMMENT CALIBRATION DISTRIBUTION, NOT HELD-OUT ACCURACY.

Feed	Network assumption	Action rate	Reach	JSD	Polarization	Largest cascade
Chronological	Matched random	0.968	0.997	0.077	0.172	0.251
Interest	Matched random	0.970	0.997	0.102	0.138	0.224
Popularity	Matched random	0.968	0.997	0.120	0.126	0.239
Chronological	Post-affiliation proxy	0.402	0.463	0.082	0.169	0.159
Interest	Post-affiliation proxy	0.400	0.463	0.097	0.163	0.161
Popularity	Post-affiliation proxy	0.400	0.461	0.115	0.164	0.159

two configured OpenRouter models. Its self-labeled prompt diagnostics do not satisfy the pre-registered linguistic, held-out correspondence, observed-cascade, or analyst-usefulness criteria. Those acceptance decisions remain open rather than being inferred from the proxy results.

VI. LIMITATIONS AND ETHICAL CONSIDERATIONS

The campaign corpus is a convenience sample of manually selected public posts from Facebook, Instagram, and TikTok and cannot represent the Thai population or each platform's users. Apify Actors can retrieve only comments exposed by their collection mechanism; the 1,000-comment request may therefore return fewer records and may reflect ranking, geography, login state, moderation, deletion, platform change, or Actor failure. Coordinated activity and bots may also bias observed response. Requested and returned counts, Actor versions, and failures must consequently accompany every result. Even lightweight normalization can alter pragmatic cues, so linguistic claims must be checked against the paired raw view.

Personas can reproduce stereotypes even when generated from real traces. DEEDY therefore avoids inferring sensitive attributes from isolated posts, reports subgroup results only where data and ethics permit, and presents regional or dialect attributes as uncertain rather than essential. Public availability does not remove privacy risk: identifiers must be hashed, raw text access restricted, quotations minimized, and collection reviewed against platform terms and institutional research-ethics requirements.

Finally, LLM agents share training data, architecture, and prompt conventions; agreement between them may reflect a common model prior rather than social consensus. Self-reported sentiment can also reflect compliance with output instructions rather than a stable latent reaction. Provider routing, model revision, quantization, load, and pricing may change latency, cost, and output even when the public model identifier remains fixed; the run manifest is therefore necessary but cannot eliminate provider-side uncertainty. Counterfactual campaign findings are conditional on the scenario, actor sampling, network, recommender, and model version. DEEDY is therefore an instrument for exploratory comparison and hypothesis generation, not a replacement for campaign experiments, customer research, surveys, or participatory research with Thai communities.

VII. Conclusion and Future Work

This work specifies DEEDY as a Thai campaign-simulation adaptation of MiroFish and implements reproducible generative and non-generative pilot paths. The repaired Docker stack served both the FastAPI backend and current React frontend; 1,000 non-generative runs and 500 OpenRouter calls completed with run-level provenance. DeepSeek V4 Flash 0731 cost less in this run and Qwen3 8B returned all requested items faster, but paired sentiment-JSD intervals did not establish a fidelity advantage for grounding, either model, or the LLM tier. The mechanism pilot exposed a separate operating boundary: propagation outcomes were far more sensitive to the assumed network than to population size, while the available crawl provided no reply links with which to validate either network or its cascades. A positive message-frame counterfactual changed simulated sentiment as designed but increased distance from the calibration distribution; it therefore demonstrates pipeline sensitivity, not campaign effectiveness.

These findings do not validate the paper's central claims about Thai linguistic fidelity or the hybrid LLM architecture. Completing that evaluation requires freezing independent scenario packets and campaign-level train/development/test splits, collecting reply- and timestamp-preserving traces, adding a matched Thai-specialized model condition, and obtaining blinded native-Thai and analyst ratings. The resulting experiments should retain the manifest and privacy controls established here, replace the post-affiliation proxy with observable trace-enriched graphs where available, and report failures whenever the pre-registered acceptance criteria are not met.

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